

Math 252 - Calculus II - Midterm Exam 2

CRN: 23526

February 28, 2025

Last Name: _____, First Name: _____

1. There are 9 problems. Many problems have multiple parts. The total credit is 100.
2. No collaboration is allowed for this exam. The work on this exam is to be yours and yours alone. Failure to adhere to this policy will result in a zero on this exam. Any instances of suspected cheating will also be reported to the Dean of Students' Office.
3. No calculators, books or other material are permitted.
4. In order to receive full credit you need to show your work, and present your answers as clearly as possible.
5. Answers must be exact (like $\sqrt{2}$ or π), not approximate (like 1.414, 3.14), unless the problem specifically indicates otherwise.
6. Simplify where appropriate. Quantities like $\sqrt{9}$, $\ln e$, inverse trigonometric functions at special values, or trigonometric functions at special angles should be calculated. Reduce all fractions.
7. If an answer line is provided for the answer, place your final answer in it. Otherwise, circle, box, or underline each final answer as appropriate for clarity.
8. Failure to follow directions specific to a problem will result in loss of points.
9. Put away and **silence** your cell phones, tablets, laptops, etc.
10. Once this exam begins, you will have 50 minutes to complete your solutions.

1. [5 pts] Set up the partial fraction decomposition of the following rational function. DO NOT determine the numerical values of the coefficients. DO NOT integrate the function.

$$\frac{x^4 + x^3 - 2x^2 + 1}{(x+1)(x-1)^2(x^2+4)^2} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2} + \frac{Dx+E}{x^2+4} + \frac{Fx+G}{(x^2+4)^2}$$

2. [5 pts] Complete the following arguments of using Comparison Test to determine if an improper integral is convergent or divergent.

The improper integral $\int_1^{\infty} \frac{\cos^2 x}{\sqrt[5]{x^6+2}} dx$ is Convergent or Divergent (circle the correct answer)

because $\frac{\cos^2 x}{\sqrt[5]{x^6+2}}$ B F (choose one between A and B, and then choose one among C, D, E, and F).

A). $\geq$ B). $\leq$, C). $\frac{1}{x^{1/5}}$ D). $\frac{1}{x^{1/6}}$ E). $\frac{1}{x^{5/6}}$ F). $\frac{1}{x^{6/5}}$

3. [15 pts] Use the method of partial fractions to evaluate $\int \frac{5x-4}{x(2x-1)} dx$.

Note: no credit will be given if you do not use partial fraction method.

$$\frac{5x-4}{x(2x-1)} = \frac{A}{x} + \frac{B}{2x-1}$$

$$5x-4 = A(2x-1) + Bx$$

$$5x-4 = 2Ax-A+Bx$$

$$5x-4 = (2A+B)x - A$$

$$\Rightarrow \begin{aligned} -A &= -4 & A &= 4 \\ 2A+B &= 5 & B &= -3 \end{aligned}$$

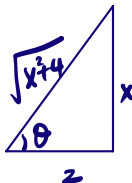
$$\int \frac{5x-4}{x(2x-1)} dx = \int \frac{4}{x} dx - \int \frac{3}{2x-1} dx = 4 \ln|x| - \frac{3}{2} \ln|2x-1| + C$$

4. [15 pts] Use trigonometric substitution to evaluate $\int \frac{1}{(\sqrt{x^2+4})^3} dx$.

Note: Simplify your answer so that it does not contain any trigonometric composed with inverse trigonometric terms.

$$\begin{aligned} x &= 2 \tan \theta \\ dx &= 2 \sec^2 \theta d\theta \end{aligned} \quad \sqrt{x^2+4} = \sqrt{4 \tan^2 \theta + 4} = \sqrt{4 \sec^2 \theta} = 2 \sec \theta$$

$$\begin{aligned} \int \frac{1}{(\sqrt{x^2+4})^3} dx &= \int \frac{1}{8 \sec^3 \theta} 2 \sec^2 \theta = \frac{1}{4} \int \frac{1}{\sec \theta} d\theta = \frac{1}{4} \int \cos \theta d\theta = \frac{1}{4} \sin \theta + C \\ &= \frac{1}{4} \cdot \frac{x}{\sqrt{x^2+4}} + C \end{aligned}$$

$$\begin{aligned} x &= 2 \tan \theta \\ \Rightarrow \frac{x}{2} &= \tan \theta \end{aligned}$$


$$\Rightarrow \sin \theta = \frac{x}{\sqrt{x^2+4}}$$

5. [12 pts] Evaluate $\int x e^{-4x} dx$. $u = x$ $dv = e^{-4x} dx$
 $du = dx$ $v = -\frac{1}{4} e^{-4x}$

$$\begin{aligned} &= -\frac{1}{4} x e^{-4x} + \int \frac{1}{4} e^{-4x} dx \\ &= -\frac{1}{4} x e^{-4x} - \frac{1}{16} e^{-4x} + C \end{aligned}$$

6. [12 pts] Find the average value of $f(x) = \tan^3 x \sec x$ on $[0, \frac{\pi}{3}]$.

$$\begin{aligned}
 & \frac{1}{\frac{\pi}{3}} \int_0^{\frac{\pi}{3}} \tan^3 \theta \sec \theta \, d\theta \\
 &= \frac{3}{\pi} \int_0^{\frac{\pi}{3}} \tan^2 \theta (\tan \theta \sec \theta) \, d\theta \\
 &= \frac{3}{\pi} \int_0^{\frac{\pi}{3}} (\sec^2 \theta - 1) \tan \theta \sec \theta \, d\theta \quad \begin{array}{l} u = \sec \theta \\ du = \sec \theta \tan \theta \, d\theta \\ \text{When } x=0, u=1 \\ x=\frac{\pi}{3}, u=2 \end{array} \\
 &= \frac{3}{\pi} \int_1^2 (u^2 - 1) \, du \\
 &= \frac{3}{\pi} \left(\frac{1}{3} u^3 - u \right) \Big|_1^2 \\
 &= \frac{3}{\pi} \left[\left(\frac{8}{3} - 2 \right) - \left(\frac{1}{3} - 1 \right) \right] = \frac{3}{\pi} \cdot \frac{4}{3} = \frac{4}{\pi}
 \end{aligned}$$

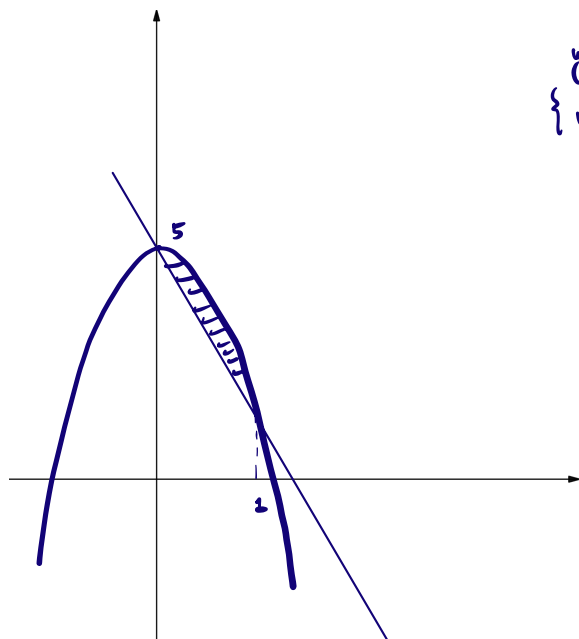
7. [12 pts] Evaluate the improper integral $\int_1^{\infty} \frac{\arctan x}{x^2 + 1} \, dx$ or show that it is divergent.

Note: Use the limit definition of improper integrals to compute the above integral. Do not use Comparison Test.

$$\begin{aligned}
 & u = \arctan x, \quad du = \frac{1}{x^2 + 1} \\
 & \int \frac{\arctan x}{x^2 + 1} \, dx = \int u \, du = \frac{1}{2} u^2 + C = \frac{1}{2} (\arctan x)^2 + C \\
 & \int_1^{\infty} \frac{\arctan x}{x^2 + 1} \, dx = \lim_{t \rightarrow \infty} \int_1^t \frac{\arctan x}{x^2 + 1} \, dx = \lim_{t \rightarrow \infty} \frac{1}{2} (\arctan x)^2 \Big|_1^t \\
 &= \lim_{t \rightarrow \infty} \frac{1}{2} (\arctan t)^2 - \frac{1}{2} (\arctan 1)^2 \\
 &= \frac{1}{2} \cdot \left(\frac{\pi}{2} \right)^2 - \frac{1}{2} \left(\frac{\pi}{4} \right)^2 \\
 &= \frac{\pi^2}{8} - \frac{\pi^2}{32} \\
 &= \frac{3\pi^2}{32}. \quad \text{Convergent.}
 \end{aligned}$$

8. [12 pts] This problem has *two* parts.

- 1). Sketch the graph of the region bounded by $y = 5 - 2x$, and $y = 5 - x^2$, and find the coordinates of the intersection points of these two curves. *Make sure to shade the region.*



$$\begin{aligned} \begin{cases} y = 5 - 2x \\ y = 5 - x^2 \end{cases} &\Rightarrow 5 - 2x = 5 - x^2 \\ &\Rightarrow x^2 - 2x = 0 \\ &\Rightarrow x(x - 2) = 0 \\ &\Rightarrow x = 0 \quad \text{or} \quad x = 2 \\ &\Rightarrow y = 5 \quad \quad \quad y = 1 \end{aligned}$$

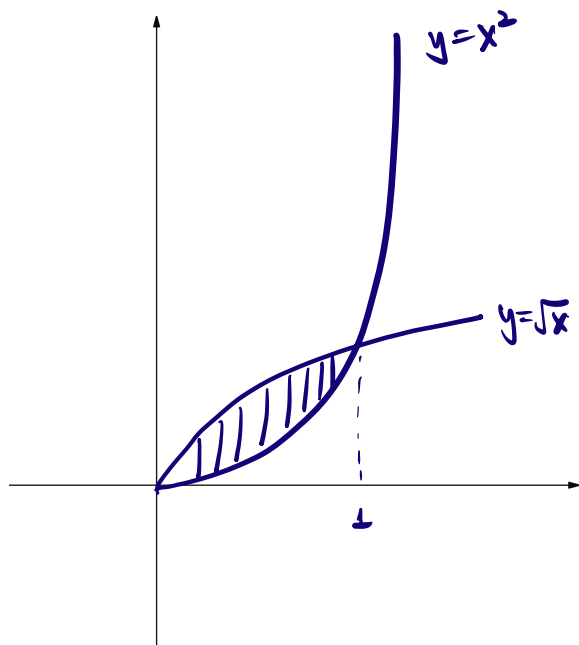
- 2). Find the **area** of the region given in Part (1).

$$\begin{aligned} \text{Area} &= \int_0^2 (5 - x^2) - (5 - 2x) \, dx \\ &= \int_0^2 (2x - x^2) \, dx \\ &= \left(x^2 - \frac{1}{3}x^3 \right) \Big|_0^2 \\ &= 4 - \frac{8}{3} \\ &= \frac{4}{3} \end{aligned}$$

Answer: Area of the region = $\frac{4}{3}$.

9. [12 pts] This problem has *two* parts.

- 1). Sketch the graph of the region bounded by $y = \sqrt{x}$ and $y = x^2$, and find the coordinates of the intersection points of these two curves. *Make sure to shade the region.*



$$x^2 = \sqrt{x}$$

$$x^4 = x$$

$$x(x^3 - 1) = 0$$

$$x=0 \text{ or } x=1$$

- 2). Use disk or washer method to find the **volume** of the solid of revolution obtained by rotating the region in Part (1) about x -axis.

$$V = \int_0^1 \pi (\sqrt{x})^2 - \pi (x^2)^2 dx$$

$$= \pi \int_0^1 (x - x^4) dx$$

$$= \pi \cdot \left(\frac{1}{2}x^2 - \frac{1}{5}x^5 \right) \Big|_0^1$$

$$= \pi \left(\frac{1}{2} - \frac{1}{5} \right)$$

$$= \pi \cdot \frac{3}{10} = \frac{3\pi}{10}$$

Answer: The volume of the solid is $\frac{3\pi}{10}$.

Formula Sheet

$$\sin^2 x + \cos^2 x = 1, \quad \sin^2 x = 1 - \cos^2 x, \quad \cos^2 x = 1 - \sin^2 x$$

$$\sec^2 x = \tan^2 x + 1, \quad \tan^2 x = \sec^2 x - 1$$

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}, \quad \sin 2x = 2 \sin x \cos x$$

$$\sin A \cos B = \frac{1}{2}[\sin(A - B) + \sin(A + B)]$$

$$\cos A \cos B = \frac{1}{2}[\cos(A + B) + \cos(A - B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\int \cot x \csc x \, dx = -\csc x + C, \quad \int \csc^2 x \, dx = -\cot x + C$$

$$\int \tan x \, dx = \ln |\sec x| + C, \quad \int \cot x \, dx = \ln |\sin x| + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C, \quad \int \csc x \, dx = \ln |\csc x - \cot x| + C$$

$$\int \sec^3 x \, dx = \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln |\sec x + \tan x| + C$$

$$\int \frac{1}{1+x^2} \, dx = \arctan x + C, \quad \int \frac{1}{\sqrt{1-x^2}} \, dx = \arcsin x + C$$

$$\int \frac{1}{a^2+x^2} \, dx = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C, \quad \int \frac{1}{\sqrt{a^2-x^2}} \, dx = \arcsin\left(\frac{x}{a}\right) + C$$

$$\int_1^\infty \frac{1}{x^p} \, dx = \begin{cases} \text{is convergent} & \text{if } p > 1 \\ \text{is divergent} & \text{if } 0 < p \leq 1. \end{cases} \quad \int_0^1 \frac{1}{x^p} \, dx = \begin{cases} \text{is convergent} & \text{if } 0 < p < 1 \\ \text{is divergent} & \text{if } p \geq 1. \end{cases}$$

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined